

Area of Shapes — Triangle & Trapezoid (with TWO Interpretations)

Student: Natalie Yoo

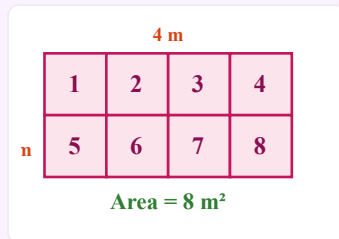
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Standard: CCSS.MATH.CONTENT.6.G.A.1 — Find the area of triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles.

1. What IS Area?

★ The big idea

Area = the number of unit squares that fit inside a shape. If the unit is meters, the squares are $1 \text{ m} \times 1 \text{ m}$, and we measure area in m^2 . If the unit is centimeters, the squares are $1 \text{ cm} \times 1 \text{ cm}$ and we measure in cm^2 .



A $4 \text{ m} \times 2 \text{ m}$ rectangle holds 8 unit squares

Count them. The rectangle has 4 columns and 2 rows of $1 \text{ m} \times 1 \text{ m}$ squares — that's **8 squares**. Each square = 1 m^2 of area, so total area = 8 m^2 .

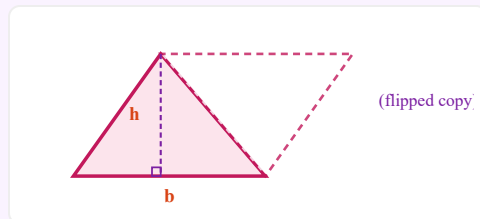
The same idea works for ANY shape. We just need a way to *count* the squares — even when the shape doesn't fit nicely on a grid (triangle, trapezoid, etc.). That's where the formulas come from.

For a rectangle: Area = $b \times h$

2. Triangle Area — Any Base Works!

★ Key insight #1 — Triangle = half of a parallelogram

Take any triangle. Make a copy, rotate it 180° and place it next to the original. The two triangles form a **parallelogram** (or rectangle) with the same base and height. So:



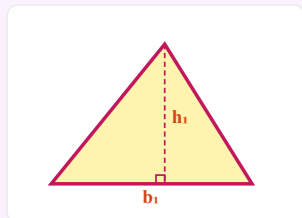
Parallelogram area = $b \times h$.

One triangle = half of that.

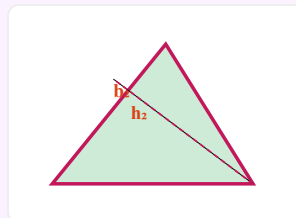
$$\text{Area}_\triangle = \frac{b \times h}{2}$$

★ Key insight #2 — ANY of the 3 sides can be the base

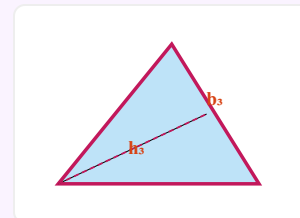
A triangle has **three sides**. ANY one of them can be called the base — but then the height **MUST** be the *perpendicular* distance from that base to the opposite vertex. The same triangle gives you **three different (b, h) pairs**, but they all multiply to the same area.



Bottom side as base



Left slant side as base



Right slant side as base

$b_1 \cdot h_1 = b_2 \cdot h_2 = b_3 \cdot h_3$ — three different multiplications, same area. Pick whichever pair is easiest to read off the figure.

Worked Example — Triangle $b = 10$ cm, $h = 4$ cm

$$\text{Area} = \frac{10 \times 4}{2} = \frac{40}{2} = 20 \text{ cm}^2$$

Important: 4 cm is the perpendicular distance from the chosen 10-cm base to the top vertex — NOT the slanted side.

3. Trapezoid Area — TWO Ways to See It

★ The formula (memorize the IDEA, not the symbols)

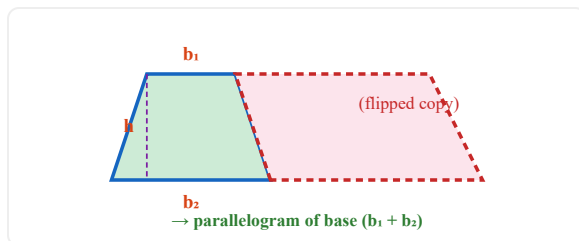
A trapezoid has two parallel sides — call them b_1 (top) and b_2 (bottom) — and a perpendicular height h between them.

$$\text{Area}_{\text{trap}} = \frac{(b_1 + b_2) \times h}{2}$$

WHY does it look like that? There are two equally good explanations. Pick the one that "clicks" for you — both prove the formula is correct.

WAY A — Half of TWO trapezoids (flip into a parallelogram)

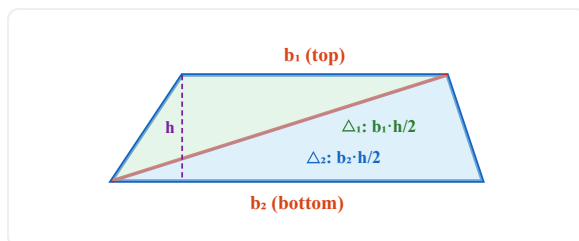
Take the trapezoid. Make an **identical copy**. Flip the copy upside down and slide it next to the original. The two trapezoids together form a **parallelogram**.



- Parallelogram area = base $\times$ height = $(b_1 + b_2) \cdot h$
- One trapezoid = HALF of that $\rightarrow$ divide by 2
- Area = $\frac{(b_1 + b_2) \cdot h}{2}$ ✓

WAY B — SUM of two triangles (cut along the diagonal)

Take the trapezoid. Draw a diagonal from one corner to the opposite corner. This cuts the trapezoid into **two triangles** sharing the same perpendicular height h — but with different bases (b_1 and b_2).



- Top triangle area = $\frac{b_1 \cdot h}{2}$
- Bottom triangle area = $\frac{b_2 \cdot h}{2}$
- Trapezoid = sum: $\frac{b_1 \cdot h}{2} + \frac{b_2 \cdot h}{2} = \frac{(b_1 + b_2) \cdot h}{2} \checkmark$

Same answer. Different picture in your head. **Both interpretations live behind that one formula.**

Worked Example — Trapezoid $b_1 = 3$, $b_2 = 6$, $h = 4$ (all in cm)

$$\text{Area} = \frac{(3 + 6) \times 4}{2} = \frac{9 \times 4}{2} = \frac{36}{2} = 18 \text{ cm}^2$$

Way A check: two trapezoids $\rightarrow 9 \times 4 = 36$ parallelogram, half = 18. $\checkmark$

Way B check: triangle $3 \cdot 4/2 = 6$ + triangle $6 \cdot 4/2 = 12$, sum = 18. $\checkmark$

★ One-Page Summary

- **Area** = count the unit squares (m^2 , cm^2 , in^2 , ...).
- **Rectangle:** Area = $b \cdot h$
- **Triangle:** Area = $\frac{b \cdot h}{2}$ (any side can be base; height must be perpendicular)
- **Trapezoid:** Area = $\frac{(b_1 + b_2) \cdot h}{2}$ (Way A: half of doubled-up parallelogram · Way B: sum of two triangles)